

Model Equations

Common conventions and units [1]

All three multicomponent models are implemented in dimensionless form. Throughout this document,

$$D_i \equiv \partial_i - iA_i, \quad D_t \equiv \partial_t + i\mu, \quad i \in \{x, y\}, \quad (1)$$

where μ is the scalar electric potential (the same quantity is sometimes denoted ϕ). Complex conjugation is denoted by an asterisk. The gauge transformation

$$\psi_n \rightarrow e^{i\chi} \psi_n, \quad \mathbf{A} \rightarrow \mathbf{A} + \nabla\chi, \quad \mu \rightarrow \mu - \partial_t\chi \quad (2)$$

leaves all of the equations below invariant.

The pyTDGL dimensional scales, in SI, are [1]

$$\tau_0 = \mu_0 \sigma \lambda^2, \quad B_0 = B_{c2} = \frac{\Phi_0}{2\pi\xi^2}, \quad A_0 = \xi B_{c2} = \frac{\Phi_0}{2\pi\xi}, \quad (3)$$

$$J_0 = \frac{4\xi B_{c2}}{\mu_0 \lambda^2}, \quad K_0 = J_0 d = \frac{4\xi B_{c2}}{\mu_0 \Lambda}, \quad V_0 = \frac{\xi J_0}{\sigma} = \frac{4\xi^2 B_{c2}}{\mu_0 \sigma \lambda^2}, \quad (4)$$

where ξ is the coherence length, λ the London penetration depth, d the film thickness, $\Lambda = \lambda^2/d$, σ the normal-state conductivity, and $\Phi_0 = h/(2e)$ the flux quantum. Solver variables are

$$\mathbf{r} = \frac{\mathbf{r}_{\text{phys}}}{\xi}, \quad t = \frac{t_{\text{phys}}}{\tau_0}, \quad \mathbf{A} = \frac{\mathbf{A}_{\text{phys}}}{A_0}, \quad \mathbf{B} = \frac{\mathbf{B}_{\text{phys}}}{B_0}, \quad \mu = \frac{V_{\text{phys}}}{V_0}, \quad \mathbf{j} = \frac{\mathbf{K}_{\text{phys}}}{K_0}. \quad (5)$$

Order parameters and code coefficients are dimensionless. The source papers use Gaussian cgs; all dimensional formulas below are converted to SI, while dimensionless ratios are unit-system independent.

For each model, current conservation is imposed through

$$\mathbf{j}_n = -\nabla\mu - \partial_t\mathbf{A}, \quad (6)$$

$$\nabla^2\mu = \nabla \cdot \mathbf{j}_s - \nabla \cdot \partial_t\mathbf{A}, \quad (7)$$

where $\mathbf{j}_s$ is the transport-normalized supercurrent specified for each model below. The multicomponent solvers presently use a prescribed vector potential; self-consistent magnetic screening is not implemented.

1 s+d [2, 3]

1.1 Order-parameter equations

Let ψ_d and ψ_s denote the dimensionless d - and s -wave order parameters. The implemented mixed-symmetry TDGL equations follow the dimensionless $d + s$ framework of Refs. [2, 3]:

$$D_t\psi_d = (D_x^2 + D_y^2)\psi_d + \varepsilon\psi_d - |\psi_d|^2\psi_d - \frac{\tau_3}{2}|\psi_s|^2\psi_d - \tau_4\psi_s^2\psi_d^* + \eta_v(D_x^2 - D_y^2)\psi_s, \quad (8)$$

$$\eta_s D_t\psi_s = \eta_s(D_x^2 + D_y^2)\psi_s + \nu\psi_s - \tau_1|\psi_s|^2\psi_s - \frac{\tau_3}{2}|\psi_d|^2\psi_s - \tau_4\psi_d^2\psi_s^* + \eta_v(D_x^2 - D_y^2)\psi_d. \quad (9)$$

Here $\varepsilon = \varepsilon(\mathbf{r}, t)$ is the solver input `disorder_epsilon`; it changes only the quadratic drive of the d sector. The standard homogeneous value is $\varepsilon = 1$.

Define the mixed directional currents

$$Q_i = \text{Im}(\psi_d^* D_i \psi_s + \psi_s^* D_i \psi_d). \quad (10)$$

The condensate current used by the implementation before electromagnetic normalization is

$$J_{\text{cond},x} = \text{Im}(\psi_d^* D_x \psi_d) + \eta_s \text{Im}(\psi_s^* D_x \psi_s) - \eta_v Q_x, \quad (11)$$

$$J_{\text{cond},y} = \text{Im}(\psi_d^* D_y \psi_d) + \eta_s \text{Im}(\psi_s^* D_y \psi_s) + \eta_v Q_y. \quad (12)$$

The transport-normalized current stored by the solver and used in Eq. (7) is

$$\mathbf{j}_s = \frac{\mathbf{J}_{\text{cond}}}{\beta_{\text{em}}}. \quad (13)$$

The opposite signs of the x and y mixed-current terms encode the $d_{x^2-y^2}$ symmetry.

1.2 Dimensional origin and SI units

From effective masses m_d^*, m_s^*, m_v^* , quadratic coefficients α_d, α_s , quartic coefficients $\beta_1, \dots, \beta_4$, diffusion coefficient $\mathcal{D}$, and conductivity σ , Gonçalves *et al.* obtain [2]

$$\eta_s = \frac{m_d^*}{m_s^*}, \quad \eta_v = \frac{m_d^*}{m_v^*}, \quad \nu = \frac{\alpha_s}{\alpha_d}, \quad (14)$$

$$\tau_1 = \frac{\beta_1}{\beta_2}, \quad \tau_3 = \frac{\beta_3}{\beta_2}, \quad \tau_4 = \frac{\beta_4}{\beta_2}, \quad (15)$$

$$\beta = \mu_0 \sigma \kappa_d^2 \mathcal{D} = \frac{\mu_0 \sigma \lambda_d^2}{\xi_d^2 / \mathcal{D}}. \quad (16)$$

β is the ratio of the electromagnetic relaxation time $\mu_0 \sigma \lambda_d^2$ to the order-parameter diffusion time $\xi_d^2 / \mathcal{D}$; Eq. (16) is the SI conversion of the paper's expression.

The remaining source scales are

$$|\psi_{d0}|^2 = \frac{\alpha_d}{2\beta_2}, \quad \xi_d^2 = \frac{\hbar^2}{2m_d^* \alpha_d}, \quad \lambda_d^2 = \frac{2m_d^* \beta_2}{\mu_0 (e^*)^2 \alpha_d}, \quad (17)$$

$$\kappa_d = \frac{\lambda_d}{\xi_d}, \quad B_{c2} = \frac{\Phi_0}{2\pi \xi_d^2}, \quad \Phi_0 = \frac{h}{2e}. \quad (18)$$

SI units are: m_i^* [kg], $\mathcal{D}$ [$\text{m}^2 \text{s}^{-1}$], σ [S m^{-1}], ξ_d, λ_d [m], B [T], $\mathbf{A}$ [Tm], and Φ_0 [Wb]. Each set (α_s, α_d) and $(\beta_1, \dots, \beta_4)$ must use common compatible units; their ratios are dimensionless for any consistent order-parameter normalization.

1.3 Parameters

N.B. Theoretical values for the parameters have been derived microscopically, corresponding to those in section VIII of Gonçalves, but we allow them to be adjusted just in case.

Code name	Meaning	Default
η_s (eta_s)	Effective-mass ratio m_d^*/m_s^* ; sets relative s -sector stiffness and the coefficient of $D_t\psi_s$.	1
η_v (eta_v)	Mixed-gradient effective-mass ratio m_d^*/m_v^* , responsible for the anisotropic s - d coupling.	0
ν (nu)	Quadratic ratio α_s/α_d ; gives the free-energy term $-\nu \psi_s ^2$.	-1
τ_1 (tau1)	Quartic ratio β_1/β_2 , comparing the s self-interaction with the d self-interaction.	1
τ_3 (tau3)	Density-coupling ratio β_3/β_2 ; TDGL cross-density coefficient $\tau_3/2$.	0
τ_4 (tau4)	Phase-sensitive quartic ratio β_4/β_2 , coupling $\psi_s^2\psi_d^*$ and $\psi_d^2\psi_s^*$.	0
β_{em} (beta_em)	SI electromagnetic factor $\beta = \mu_0\sigma\kappa_d^2\mathcal{D}$. Since the solver prescribes $\mathbf{A}$ rather than evolving the paper's $\beta\partial_t\mathbf{A}$ equation, this field acts only as the current divisor in Eq. (13).	1
ε (disorder_epsilon)	Local d -sector quadratic coefficient, interpretable as $\alpha_d(\mathbf{r},t)/\alpha_{d,\text{ref}}$; the homogeneous Gonçalves normalization gives $\varepsilon = 1$.	1

The validation conditions are

$$\eta_s > 0, \quad \eta_v > \eta_v^2, \quad \tau_1 > 0, \quad \beta_{\text{em}} > 0, \quad (19)$$

with $\varepsilon \leq 1$ enforced by the solver. The condition $\eta_s > \eta_v^2$ makes the mixed-gradient sector positive definite.

1.4 Limitations and approximations

The Gonçalves formulation treats the diffusion coefficient and conductivity phenomenologically and does not impose a strict clean- or dirty-limit condition. The microscopic theory of Zhu *et al.* includes magnetic and nonmagnetic impurity scattering rather than assuming only the asymptotic dirty limit. Nevertheless, the implementation's local Ohmic normal current and diffusive relaxation are dirty-limit-like; quantitative dynamics require material-specific scattering and relaxation parameters.

The field $\varepsilon(\mathbf{r},t)$ changes only the d -sector quadratic coefficient and is not a complete microscopic disorder model. The factor β_{em} is only the current normalization in Eq. (13). A scalar terminal value fixes ψ_d and sets $\psi_s = 0$ at terminal sites, which is a numerical contact model rather than a general $s + d$ interface condition.

Note that H_{c2} here is for a pure d-wave and therefore the “actual” H_{c2} could be different (a possible simulation result).

2 d+d' [4, 5]

2.1 Free energy and equilibrium-relaxation equations

Let d denote the dominant $d_{x^2-y^2}$ component and d' the subdominant d_{xy} component. In the code they are stored as $\psi_1 = d$ and $\psi_2 = d'$, respectively. The dimensionless free-energy density of Lei, Aruna, and Wang [4] is augmented by the optional orbital-Zeeman coupling derived microscopically by Wang and Wang [5]:

$$\begin{aligned}
f_{d+d'} = & -|d|^2 - \alpha|d'|^2 + \frac{1}{2}(|d|^4 + |d'|^4) + \frac{1}{3}|d|^2|d'|^2 + \frac{1}{6}(d^*d' + dd'^*)^2 \\
& + |\mathbf{D}d|^2 + |\mathbf{D}d'|^2 + \kappa^2 b^2 - ig_Z b(d^*d' - dd'^*), \quad (20)
\end{aligned}$$

where

$$b = (\nabla \times \mathbf{A})_z = \frac{B_z}{B_{c2}}, \quad B_{c2} = \frac{\Phi_0}{2\pi\xi^2}. \quad (21)$$

The original field-driven model in Ref. [4] deliberately omits the last term; setting $g_Z = 0$ reproduces that functional. The signed coefficient g_Z incorporates the microscopic particle-hole asymmetry and the dimensionless normalization of the orbital-Zeeman interaction. The code computes b at mesh sites by a discrete Stokes curl of the edge-centered applied vector potential.

The source papers specify a static GL functional rather than physical TDGL kinetics. The implemented equations are therefore phenomenological negative gradient flow for finding stationary states:

$$r_d D_t d = (D_x^2 + D_y^2)d + d - |d|^2 d - \frac{2}{3}|d'|^2 d - \frac{1}{3}d'^2 d^* + ig_Z b d', \quad (22)$$

$$r_{d'} D_t d' = (D_x^2 + D_y^2)d' + \alpha d' - |d'|^2 d' - \frac{2}{3}|d|^2 d' - \frac{1}{3}d^2 d'^* - ig_Z b d. \quad (23)$$

The factors $2/3$ and $1/3$ follow by differentiating both quartic cross terms in Eq. (20); they are not independent fitting parameters.

At $b = 0$ and $g_Z = 0$, a uniform pure- d stationary state has $|d| = 1$ and $d' = 0$. For $1/3 < \alpha < 1$, the uniform mixed minimum instead has

$$|d|^2 = \frac{3(3-\alpha)}{8}, \quad |d'|^2 = \frac{3(3\alpha-1)}{8}, \quad \arg(d') - \arg(d) = \pm \frac{\pi}{2}. \quad (24)$$

For $g_Z b > 0$, the Zeeman term lifts this chiral degeneracy and favors $d'/d = -i|d'|/|d|$; reversing the field reverses the favored chirality.

2.2 Current and orbital magnetization

Because Eq. (20) has equal diagonal stiffnesses and no mixed-gradient term, the implemented condensate and transport currents are

$$\mathbf{J}_{\text{cond}} = \text{Im}(d^* \mathbf{D}d) + \text{Im}(d'^* \mathbf{D}d'), \quad (25)$$

$$\mathbf{j}_s = \frac{\mathbf{J}_{\text{cond}}}{c_{\text{em}}}, \quad c_{\text{em}} \equiv \text{em_coupling}. \quad (26)$$

The local dimensionless orbital magnetization associated with the Zeeman term is

$$m_z = -\left. \frac{\partial f_Z}{\partial b} \right|_{d,d'} = ig_Z (d^* d' - dd'^*), \quad f_Z = -ig_Z b (d^* d' - dd'^*), \quad (27)$$

This is the order-parameter orbital contribution and excludes the derivative of the separately prescribed $\kappa^2 b^2$ field energy. Equation (27) is exposed as the diagnostic `Solution.orbital_magnetization`.

The discrete condensate free-energy diagnostic includes the local potential, gradient energy, and orbital-Zeeman energy in Eq. (20). It omits $\kappa^2 b^2$, which is constant with respect to d, d' when $\mathbf{A}$ is prescribed and therefore does not affect order-parameter relaxation.

2.3 Parameters

Code name	Meaning	Default
α (<code>alpha</code>)	Ratio of the subdominant and dominant quadratic GL coefficients in the normalization of Ref. [4]; controls the zero-field d_{xy} instability.	0.5
r_d (<code>relaxation_d</code>)	Positive phenomenological relaxation coefficient for the dominant $d_{x^2-y^2}$ component.	1
$r_{d'}$ (<code>relaxation_d_prime</code>)	Positive phenomenological relaxation coefficient for the subdominant d_{xy} component.	1
c_{em} (<code>em_coupling</code>)	Implementation-specific conversion from condensate-current normalization to the transport normalization used by the Poisson solver.	1
g_Z (<code>zeeman_coupling</code>)	Signed dimensionless orbital-Zeeman coefficient; zero selects the unmodified Lei–Aruna–Wang model.	0

All parameters must be finite, $r_d, r_{d'}, c_{\text{em}} > 0$, and the current implementation requires $|g_Z| < 1$. Negative α is permitted. Values $\alpha > 1$ are numerically allowed with a warning because they leave the assumed subdominant- d_{xy} regime of the source model.

2.4 Limitations and approximations

Our equations are weak-coupling microscopic results based on BCS, Gor’kov, and path-integral methods without impurity scattering; they do not assume the dirty limit. The coefficients $r_d, r_{d'}$ are phenomenological additions, so the simulated trajectory is equilibrium relaxation rather than a microscopic clean-limit TDGL prediction.

The Zeeman term is optional because it changes the phase diagram qualitatively: when $g_Z b \neq 0$, it acts as a linear source for the opposite component and removes the exact degeneracy of the two chiralities. Comparisons with the phase boundary of Ref. [4] must use $g_Z = 0$; nonzero values instead implement the microscopic extension of Ref. [5].

The model has equal component stiffnesses, no mixed-gradient term, and fixed quadratic coefficients, so nonuniform `disorder_epsilon` is unsupported. The reported orbital magnetization omits its bound current from the transport Poisson solve. Transport terminals require `terminal_psi=None` and natural order-parameter boundary conditions.

3 s+s [6, 7]

3.1 Free energy and order-parameter equations

Let ψ_1 and ψ_2 denote the two dimensionless isotropic s -wave band order parameters. With $k_1 = 1$ and $k_2 = k_{2/1} \equiv \text{k2_over_k1}$, the dimensionless condensate free-energy density follows Refs. [6, 7]:

$$f = \sum_{n=1}^2 \left[a_n |\psi_n|^2 + \frac{b_n}{2} |\psi_n|^4 + k_n |\mathbf{D}\psi_n|^2 \right] - \gamma_{12} (\psi_1 \psi_2^* + \psi_2 \psi_1^*). \quad (28)$$

The implemented dissipative extension is the negative gradient flow of Eq. (28):

$$r_1 D_t \psi_1 = (D_x^2 + D_y^2) \psi_1 - a_1 \psi_1 - b_1 |\psi_1|^2 \psi_1 + \gamma_{12} \psi_2, \quad (29)$$

$$r_2 D_t \psi_2 = k_{2/1} (D_x^2 + D_y^2) \psi_2 - a_2 \psi_2 - b_2 |\psi_2|^2 \psi_2 + \gamma_{12} \psi_1. \quad (30)$$

The unscaled condensate current and the transport-normalized current are

$$\mathbf{J}_{\text{cond}} = \text{Im}(\psi_1^* \mathbf{D} \psi_1) + k_{2/1} \text{Im}(\psi_2^* \mathbf{D} \psi_2), \quad (31)$$

$$\mathbf{j}_s = \frac{\mathbf{J}_{\text{cond}}}{c_{\text{em}}}, \quad c_{\text{em}} \equiv \text{em_coupling}. \quad (32)$$

A positive γ_{12} minimizes the energy for equal component phases; a negative value favors a relative phase of π .

3.2 Dimensional origin and SI units

The Kogan–Schmalian two-band functional, converted from Gaussian cgs to SI, is

$$F_{\text{SI}} = \int dV \left\{ \sum_{\nu=1}^2 \left[a_{\nu}^{(G)} |\Delta_{\nu}|^2 + \frac{b_{\nu}^{(G)}}{2} |\Delta_{\nu}|^4 + K_{\nu}^{(G)} |\mathbf{\Pi} \Delta_{\nu}|^2 \right] - \gamma^{(G)} (\Delta_1 \Delta_2^* + \Delta_2 \Delta_1^*) + \frac{B^2}{2\mu_0} \right\}, \quad (33)$$

where $\mathbf{\Pi} = \nabla + 2\pi i \mathbf{A} / \Phi_0$, $\Phi_0 = h/(2e)$, and Δ_{ν} is the dimensional band gap. Superscript (G) labels dimensional GL coefficients, not powers.

For the clean, isotropic weak-coupling model, the paper gives [7]

$$a_{\nu}^{(G)} = \frac{N(0)}{\eta_{\lambda}} (S_{\nu} - n_{\nu} \eta_{\lambda} \tau), \quad b_{\nu}^{(G)} = \frac{N(0)}{W^2} n_{\nu}, \quad (34)$$

$$\gamma^{(G)} = \frac{N(0)}{\eta_{\lambda}} \lambda_{12}, \quad K_{\nu}^{(G)} = \frac{N(0) \hbar^2 v_{\nu}^2}{6W^2} n_{\nu}, \quad (35)$$

$$\eta_{\lambda} = \lambda_{11} \lambda_{22} - \lambda_{12}^2, \quad S_1 = \lambda_{22} - n_1 \eta_{\lambda} S, \quad S_2 = \lambda_{11} - n_2 \eta_{\lambda} S, \quad (36)$$

$$W^2 = \frac{8\pi^2 (k_B T_c)^2}{7\zeta(3)}, \quad S = \ln \left(\frac{2e^{\gamma_E} \hbar \omega_D}{\pi k_B T_c} \right), \quad \tau = 1 - \frac{T}{T_c}. \quad (37)$$

Here $N(0)$ is the total single-spin density of states at the Fermi level; $n_{\nu} = N_{\nu}/N(0)$ the band fraction; $\lambda_{\nu\mu}$ the dimensionless pairing matrix; v_{ν} the Fermi velocity; ω_D the cutoff frequency; and γ_E Euler's constant. Equations (35) are one weak-coupling realization; Eq. (33) also applies phenomenologically.

SI units are: Δ_{ν} , W , $k_B T_c$ [J], $N(0)$ [$\text{J}^{-1} \text{m}^{-3}$], $a_{\nu}^{(G)}$, $\gamma^{(G)}$ [$\text{J}^{-1} \text{m}^{-3}$], $b_{\nu}^{(G)}$ [$\text{J}^{-3} \text{m}^{-3}$], $K_{\nu}^{(G)}$ [$\text{J}^{-1} \text{m}^{-1}$], v_{ν} [m s^{-1}], $\mathbf{A}$ [T m], B [T], and F_{SI} [J]. n_{ν} , $\lambda_{\nu\mu}$, η_{λ} , S , τ are dimensionless.

With gap scale Δ_0 , length ξ , and energy-density scale $f_0 = K_1^{(G)} \Delta_0^2 / \xi^2$, the code coefficients are

$$a_{\nu} = \frac{a_{\nu}^{(G)} \xi^2}{K_1^{(G)}}, \quad b_{\nu} = \frac{b_{\nu}^{(G)} \Delta_0^2 \xi^2}{K_1^{(G)}}, \quad (38)$$

$$k_{2/1} = \frac{K_2^{(G)}}{K_1^{(G)}}, \quad \gamma_{12} = \frac{\gamma^{(G)} \xi^2}{K_1^{(G)}}. \quad (39)$$

Other scales are equivalent if the whole functional is rescaled consistently.

3.3 Parameters

Code name	Meaning	Default
a_1 (a1), a_2 (a2)	Dimensionless $a_\nu^{(G)}$; contain a constant interband contribution and a term proportional to $1 - T/T_c$.	$-1, -1$
b_1 (b1), b_2 (b2)	Dimensionless quartic self-interactions corresponding to $b_\nu^{(G)} = N(0)n_\nu/W^2$.	$1, 1$
$k_{2/1}$ (k2_over_k1)	Stiffness ratio $K_2^{(G)}/K_1^{(G)}$; depends on band fractions and squared Fermi velocities.	1
γ_{12} (josephson_gamma)	Dimensionless interband Josephson coupling corresponding to $\gamma^{(G)} = N(0)\lambda_{12}/\eta_\lambda$.	0
r_1 (relaxation1), r_2 (relaxation2)	Phenomenological TDGL kinetic coefficients; absent from the static Kogan–Schmalian functional.	$1, 1$
c_{em} (em_coupling)	Implementation-specific conversion from condensate-current normalization to the transport normalization used by the Poisson solver; it is not a coefficient in the static source functional.	1

The validation conditions are

$$b_1 > 0, \quad b_2 > 0, \quad k_{2/1} > 0, \quad r_1 > 0, \quad r_2 > 0, \quad c_{\text{em}} > 0. \quad (40)$$

a_1, a_2 may have either sign; an uncoupled uniform band with $a_n < 0$ has $|\psi_n| = \sqrt{-a_n/b_n}$.

3.4 Limitations and approximations

The microscopic coefficient map in Ref. [7] is derived for a clean, isotropic, weak-coupling material, not the dirty limit. The GL result that the two gaps are proportional near the common T_c is more general and also has dirty-limit derivations, but those dirty microscopic coefficients are not encoded here. The relaxation coefficients r_1, r_2 remain phenomenological.

The fixed-temperature coefficients a_1, a_2 require **disorder_epsilon**= 1, and the model omits interband gradient drag and impurity-induced interband scattering. Near T_c , the two stored fields do not imply two independent GL coherence lengths. The factor c_{em} is a current normalization, and scalar terminal order-parameter values are unsupported because their two-band mapping is unspecified.

References

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